

Robust and Scalable Bayesian Online Changepoint Detection

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Abstract

This paper proposes an online, provably robust, and scalable Bayesian approach for changepoint detection. The resulting algorithm has key advantages over previous work: it provides provable robustness by leveraging the generalised Bayesian perspective, and also addresses the scalability issues of previous attempts. Specifically, the proposed generalised Bayesian formalism leads to conjugate posteriors whose parameters are available in closed form by leveraging diffusion score matching. The resulting algorithm is exact, can be updated through simple algebra, and is more than 10 times faster than its closest competitor.

1. Introduction

Changepoint (CP) detection is the task of identifying sudden changes in the statistical properties of a data stream. The methods to detect CPs are used in applications including systems health monitoring (Stival et al., 2022; Yang et al., 2006), financial data (Kim et al., 2022; Kummerfeld & Danks, 2013), climate change (Reeves et al., 2007; Itoh & Kurths, 2010), and cyber security (Hallgren et al., 2022). Existing approaches include likelihood ratio methods such as the parametric method CUSUM (Page, 1954) or Change Finder methods (Kawahara & Sugiyama, 2009), to Bayesian methods such as in Chib (1998); Fearnhead (2006).

Detecting CPs in an online fashion is an even more challenging task, but can allow practitioners to act on these systems in real-time. In a Bayesian context, the most popular method is *Bayesian online changepoint detection (BOCD)* (Adams & MacKay, 2007; Fearnhead & Liu, 2007). Here, the data stream is assumed to come from one of several different underlying distributions; and the goal is to quantify our

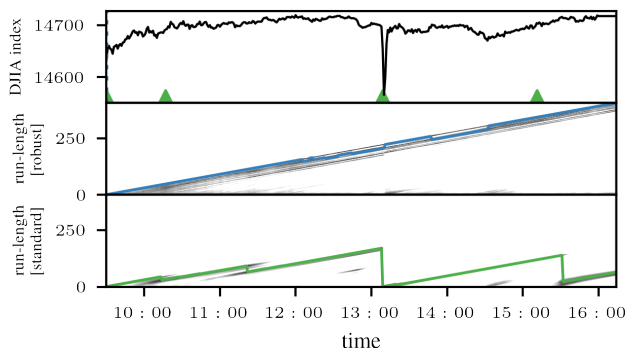


Figure 1. Twitter Flash Crash. The run-length is the time since the last changepoint (CP). *Top*: Dow Jones Index with Maximum a posteriori CPs detected by standard BOCD marked as $\blacktriangle$. *Middle & Bottom*: run-length posteriors of $\mathcal{D}_m$ -BOCD with most likely run-length in blue and of standard BOCD in green. Standard BOCD incorrectly detects a CP, $\mathcal{D}_m$ -BOCD does not.

uncertainty over the most recent time at which the data distribution changed. BOCD has many desirable properties: it is suitable for multivariate data and has the capacity to quantify uncertainty. However, it also has a significant flaw inherited from Bayesian inference: it is not robust under outliers or model misspecification. This can lead to failures, where most data points inferred to be CPs are simply mild heterogeneities in the data. This is a significant problem, and can cause practitioners to act on safety-critical systems based upon an erroneously declared CPs.

The lack of robustness in Bayesian methods has recently come to the forefront, and various strategies have been proposed to address it. Arguably the most successful amongst these have been generalised Bayesian methods (see e.g. Bisiri et al., 2016; Jewson et al., 2018; Knoblauch et al., 2022). Building on these ideas, Knoblauch et al. (2018) introduced the first robust version of BOCD using generalised Bayesian inference based on β -divergences (β -BOCD).

While the resulting algorithm is generally applicable and provides robustness, it has a major drawback that has severely impeded its broader use: it is not scalable. This is mainly due to the intractability of the generalised posterior and predictive distributions, which require multiple variational approximations to be performed at each time point. As a result, β -BOCD is practically infeasible if one is interested in online methods for high-frequency data, or if one deals with a constrained computational budget.

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This paper proposes a new generalised Bayesian inference scheme based on *diffusion score matching* (Barp et al., 2019), which is effectively a weighted version of the original score-matching divergence of Hyvärinen (2006). If the weights are chosen appropriately, the resulting posteriors are provably robust, and the corresponding CP detection algorithm, denoted $\mathcal{D}_m$ -BOCD, is also robust to outliers. This is illustrated in Figure 1 on the value of the Dow Jones Industrial Average (DJIA) on the day of the ‘Twitter flash crash’ on 17/04/2013: standard BOCD falsely identifies 3 CPs, whilst $\mathcal{D}_m$ -BOCD correctly identifies no CPs.

Additionally—and unlike posteriors based on the β -divergence— $\mathcal{D}_m$ -posteriors also have a conjugacy property for likelihoods of the exponential family so long as the prior is chosen to be a normal, truncated normal, or any other squared exponential distribution. This makes $\mathcal{D}_m$ -BOCD very fast: specifically, it ensures that all posteriors used in the algorithm can be updated exactly and efficiently through elementary vector and matrix calculations. If one uses the pruning strategies for the CP posterior proposed in Adams & MacKay (2007), the computational complexity of our algorithm is $\mathcal{O}(T(d^2 + p^2))$; where T is the length of the data stream, d is the dimension of the observations, and p is the number of model parameters. This is the same computational complexity as the original BOCD algorithm. This also makes $\mathcal{D}_m$ -BOCD more than 10 times faster than β -BOCD in our numerical experiments.

Beyond that, $\mathcal{D}_m$ -BOCD has benefits that make it more attractive than standard BOCD even from a purely computational point of view in certain settings. For example, when modelling d -dimensional observations with non-Gaussian exponential family distributions, we can obtain conjugate $\mathcal{D}_m$ -posteriors, even though no conjugate posteriors exist in the standard Bayesian case.

In summary, we make two key contributions:

- (1) We derive and propose the $\mathcal{D}_m$ -posterior; *proving its robustness and closed form updates* in the process;
- (2) We use this posterior for BOCD, leading to the first algorithm that is *both robust and scalable*.

The remainder of the paper is structured as follows: Section 2 reviews BOCD and generalised Bayesian inference. Section 3 derives the robustness and scalability properties of $\mathcal{D}_m$ -posteriors, and integrates them with BOCD. We then validate our approach experimentally in Section 4.

2. Background

Our method merges generalised Bayesian posteriors based on diffusion score matching with the BOCD algorithm. Here, we provide a short explanation of the concepts relevant for understanding this interface.

2.1. Bayesian Online Changepoint Detection (BOCD)

Let $x_{1:T}$ be a sequence of observations $x_1, x_2, \dots, x_T$, where $x_t \in \mathcal{X} \subseteq \mathbb{R}^d$ for the time index $t \in \{1, \dots, T\}$. Throughout, $x_{1:T}$ follows the product partition model of Barry & Hartigan (1993): the data is partitioned through a sequence of changepoints (CPs) $0 = \tau_1 < \tau_2 < \dots$ so that the i -th segment is $x_{\tau_i:\tau_{i+1}-1}$, and data within the i -th segment is independently and identically distributed (i.i.d.) conditional on τ_i, τ_{i+1} . In the model underlying BOCD, the data in each segment is modelled with the same model class $\{p_\theta : \theta \in \Theta\}$, but with a different parameter for each segment. The key insight for this model, reached independently by both Adams & MacKay (2007) and Fearnhead & Liu (2007), is that Bayesian inference can be made online and efficiently if, at time t , one only tracks a posterior distribution over the most recent CP. Instead of defining a prior and posterior over the CPs directly, BOCD therefore seeks to infer the so-called *run-length* r_t of the current segment—the amount of time since the most recent CP.

The remainder of this section details the hierarchical Bayesian model underlying the BOCD construction. Firstly, the approach uses a conditional prior on the run-length:

$$r_t | r_{t-1} \sim H(r_t | r_{t-1}). \quad (\text{Conditional prior on run-length})$$

Since at time t we either have a new CP ($r_t = 0$) or the current segment continues ($r_t = r_{t-1} + 1$), $H(r_t | r_{t-1})$ has positive probability mass only for $r_t \in \{0, r_{t-1} + 1\}$. See Wilson et al. (2010) for a broader discussion of prior selection. Conditional on r_t , all data points $x_{t'}$ from the same segment $(t - r_t) : t$ so that $t' \in \{t - r_t, t - r_t + 1, \dots, t\}$ are then modelled as i.i.d. from p_θ via

$$\begin{aligned} \theta &\sim \pi(\theta) && (\text{Parameter prior}), \\ x_{t'} | \theta &\sim p_\theta(x_{t'}) && (\text{Probability model for data}). \end{aligned}$$

The quantity of interest is the posterior over r_t , which is

$$p(r_t | x_{1:t}) = \frac{p(r_t, x_{1:t})}{p(x_{1:t})} = \frac{p(r_t, x_{1:t})}{\sum_{r_t=0}^t p(r_t, x_{1:t})}.$$

This shows that the run-length posterior is tractable whenever the joint distribution between run-length and observations given by $p(r_t, x_{1:t})$ is also tractable. Intriguingly, these terms can be computed efficiently via an online recursion whenever the posterior predictive is tractable:

$$p(r_t, x_{1:t}) = \sum_{r_{t-1}=0}^{t-1} \underbrace{p(x_t | x_{t-1}^{(r_t)})}_{\text{Predictive Posterior}} \underbrace{H(r_t | r_{t-1})}_{\text{CP prior}} p(r_{t-1}, x_{1:t-1}),$$

where $x_{t-1}^{(r_t)} = x_{t-r_t:t-1}$ is the segment with run-length r_t except the most recent observation x_t , and the predictive of x_t constructed from $x_{t-1}^{(r_t)}$ is

$$p(x_t | x_{t-1}^{(r_t)}) = \int_{\Theta} p_\theta(x_t) \pi^B(\theta | x_{t-1}^{(r_t)}) d\theta, \quad (1)$$

where $\pi^B(\theta|x_{t-1}^{(r_t)}) \propto \prod_{i=1}^{r_t} p_\theta(x_{t-i})\pi(\theta)$ is the Bayes posterior over θ in the current segment. To ensure that this integral is tractable in closed form, BOCD algorithms usually use prior densities $\pi(\theta)$ and models $p_\theta(x)$ forming a conjugate likelihood-prior pair.

Since the standard BOCD method was proposed, it has been extended in a wide range of directions. A full literature review is beyond the scope of this paper, but we highlight extensions to Gaussian processes models (Saatçi et al., 2010), non-exponential families (Turner et al., 2013), multiple models in different segments (Knoblauch & Damoulas, 2018; Knoblauch et al., 2022), observations with multiple fidelity levels (Gundersen et al., 2021), and prediction (Agudelo-España et al., 2020). We also note that while BOCD only quantifies uncertainty about the most recent CP, an efficient maximum a-posteriori Viterbi-style recursion can be used to efficiently update point estimates of all CP locations (see e.g. Fearnhead & Liu, 2007).

Unfortunately, BOCD is not robust: it finds spurious CPs whenever the model is a poor description of data. To address this issue, one can replace the standard Bayesian parameter posterior in (1) with a robust generalised Bayesian posterior.

2.2. Generalised Bayesian (GB) inference

If the statistical model p_θ is well-specified so that for some $\theta_0 \in \Theta$, the true data-generating mechanism is p_{θ_0} , standard Bayesian updating is the optimal way of integrating prior information with data (Zellner, 1988). Crucially, this no longer holds if the model is misspecified. In this setting, uncertainties are miscalibrated, posterior inferences are sensitive to outliers and heterogeneity, and the Bayesian update may no longer be the best way of processing information. To address these issues, a recent line of research has advocated for the use of generalised Bayesian inference (see e.g. Grünwald, 2012; Bissiri et al., 2016; Jewson et al., 2018; Knoblauch et al., 2022; Fong et al., 2021; Jewson & Rossell, 2022; Matsubara et al., 2022b) which, once conditioned on some data $x_{1:T}$, is based on a belief distribution of the form

$$\pi_\omega^\mathcal{D}(\theta|x_{1:T}) \propto \pi(\theta) \exp\{-\omega T \cdot \widehat{\mathcal{D}}(\theta)\}. \quad (2)$$

While $\widehat{\mathcal{D}}(\theta)$ could in principle represent any loss function, we consider a narrowed scope. Specifically, for $\mathcal{D}$ being a discrepancy measure on the space of probability measures on $\mathcal{X}$, and p_0 being the true data-generating process, $\widehat{\mathcal{D}} : \Theta \rightarrow \mathbb{R}$ uses $x_{1:T}$ to estimate the part of the discrepancy $\mathcal{D}(p_0, p_\theta)$ that depends on θ . Here, $\omega > 0$ is called the *learning rate* and acts as a scaling parameter that determines how quickly the posterior learns from the data. While the choice of ω may depend on various other considerations (Grünwald, 2012; Holmes & Walker, 2017), it is typically chosen to provide approximate frequentist coverage (Lyndon et al., 2019; Martin & Syring, 2022). Neither of these

techniques are suitable for the online setting; and we will therefore propose a new way of choosing ω in Section 3.4.

The posteriors in (2) are called generalised posteriors because for $\omega = 1$, and $\widehat{\mathcal{D}}(\theta) = \frac{1}{T} \sum_{t=1}^T -\log p(x_t|\theta)$ estimating the Kullback-Leibler divergence between the model and the data-generating process, one recovers the standard Bayes posterior. Using such generalisations is usually done for two main arguments: to provide robustness, and to improve computation. For example, Chernozhukov & Hong (2003) are the first to suggest them for estimation when computing a minimum is hard. Rather than focusing on computational aspects, Hooker & Vidyashankar (2014), Ghosh & Basu (2016) and Bissiri et al. (2016) advocated for their use to improve robustness. This has led to a flurry of papers proposing particular discrepancy measures that induce robustness (e.g. Chérif-Abdellatif & Alquier, 2020), and their various applications in sequential Monte Carlo (Boustati et al., 2020), deep Gaussian processes (Knoblauch, 2019), and Bayesian neural networks (Futami et al., 2018). More recently, a line of work has exploited generalised posteriors both for computational gain and robustness: Matsubara et al. (2022b;a) showcased their use for robust inference in unnormalised models with both continuous and discrete data. Similarly, Schmon et al. (2020); Dellaporta et al. (2022); Pacchiardi & Dutta (2021); Legramanti et al. (2022) have used them for robustness in simulation-based and likelihood-free settings.

2.3. Generalised Bayesian Inference in BOCD

Knoblauch et al. (2018) first proposed a robustification of BOCD based on (2) and the β -divergence, which is robust and well-defined for $\beta \in (0, \infty)$ when p_θ is uniformly bounded on $\mathcal{X}$, and whose natural estimator was derived by Basu et al. (1998) and is given by

$$\widehat{\mathcal{D}}_\beta(\theta) = \frac{1}{T} \sum_{t=1}^T \frac{1}{1+\beta} \int_{\mathcal{X}} p_\theta(x)^{1+\beta} dx + \frac{1}{\beta} p_\theta(x_t)^\beta.$$

While the resulting method can be made robust, it has several key failures that make it computationally infeasible in most settings. Firstly, the loss depends on $\int_{\mathcal{X}} p_\theta(x)^{1+\beta} dx$. Unless this integral is available in closed form, using $\widehat{\mathcal{D}}_\beta$ will introduce the same challenges as working with an intractable likelihood in a standard Bayesian setting. Secondly, the hyperparameter β enters the loss as the exponent of a likelihood. Numerically, this makes the loss extremely sensitive to even very minor changes in β , which makes it very difficult to tune β and counteracts the very robustness one hopes to achieve. This numerical instability is compounded by the fact that (2) depends on the exponentiation of $\widehat{\mathcal{D}}_\beta$ —if p_θ is an exponential family member, then even if one ignores the integral term, $\exp\{-\omega T \widehat{\mathcal{D}}_\beta(\theta)\}$ is a double exponential. Thirdly, posteriors based on $\widehat{\mathcal{D}}_\beta$ often have to be approximated using variational methods. Since this has to